Gadget Recommendation System with Large Language Model using TAG

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***Abstract*—In this fast-moving world of digital media, consumer electronics make it difficult and time-consuming for users to choose the best gadget. To overcome the problem, we propose a Gadget Recommendation System empowered by Large Language models, using Llama-3 8B, to help provide personalized recommendations for various gadgets, including mobiles and laptops. The system primarily uses the Table-Augmented Generation (TAG) approach, which allows a structured language combined with the LLaMA-3 8 Billion tokens, enabling the chatbot to provide accurate and context-specific suggestions. Users communicate with the system by asking gadget-related questions translated into SQL queries through TAG's Query Synthesis process. This information is then extracted from the SQL database and compared to the query to provide LLaMA with the right and relevant answer [1]. Using TAG, which combines structured data with language modeling techniques, the system thus increases user satisfaction and helps make better decisions by providing relevant gadget suggestions. This approach demonstrates the possibilities of applying structured data with LLM for large-scale, efficient, and customer-oriented recommendation systems in electronics.**

I. Introduction and Statement of the Problem

In the current digital world, the consumer electronics market is growing at a fast pace, users are offered a vast number of choices in choosing the gadgets they need, for example, smartphones, laptops, and other gadgets. Decision-making in this environment is typically challenging because of the features users demand for gadgets that offer exact solutions to their requirements and tastes. In response to this challenge, most recommendation systems are deployed to help reduce the time required to filter out a certain gadget through recommendations from individual user input. One of the drawbacks of previous conventional recommendation systems including the collaborative filtering approach or content-based systems is unable to comprehend the interpretations of the context and inquiries based on complex user necessities and personal preferences. New changes in the Large Language Models (LLMs), especially LLaMA-3 8 Billion have an excellent solution in handling natural language search queries. Our project also develops a Personalized Gadget Recommendation System consisting of two components, LLaMA-3 (8 Billion tokens) for natural language processing and a MySQL database for structured information. The system utilizes the Table-Augmented Generation (TAG) method that combines structured data querying with a Large language model, to achieve better accuracy and relevance of recommendations [2].

In our approach in TAG's Query Synthesis, the system translates the user query into SQL, the system searches for a relevant set of data and then sends it to LLaMA to provide the user with a recommendation in natural language [1]. Thus, instead of a simple Text2SQL process, the TAG-driven procedure enables elaborate interconnections between the structured data and the language models. In this project, by using advanced AI techniques we aim to generate responses in natural language enhancing user satisfaction with gadget recommendations.

II. Review of Literature

1. Text2SQL is Not Enough: Unifying AI and Databases with TAG" [1]: This paper focuses on the Table-Augmented Generation (TAG) model where the concern lies in bridging databases with structures to advanced large language models (LLMs) of the natural language query interfaces. This is exemplified by current systems such as Text2SQL which are only capable of transforming natural language queries into query language for pulling information from structured data. To achieve these, the TAG model breaks its function into three steps namely query synthesis, query execution, and answer generation where user input is in the right from SQL, effectively executed in the respective database then processed by LLM to return meaningful responses to users. Specifically, this article is closely related to the work done for our Gadget Recommendation System, where, in addition to LLMs such as LLAMA, we employ a MySQL database to offer tailored

gadget recommendations. Thus, by following the TAG-like strategy, our system can perform a more effective natural language understanding of the queries and better functionality of the database search for particular data, as well as generate recommendations based on the results of the search. The use of combining structured querying along with LLM-based reasoning, as described in the article of choice, will improve the efficiency and effectiveness of suggestions on the gadgets by making the kind of system smoother to operate and more responsive to the user's needs. The capacity of the TAG framework for iterative reasoning also guarantees the handling of more extended queries, meeting the aims of the project.

2. Evaluating the Efficacy of Open-Source LLMs in Enterprise-Specific RAG Systems: A Comparative Study of Performance and Scalability [2]: This paper discusses the methods employed in combining artificial intelligence models and database systems for enhancing the analysis and processing of natural language queries, in related discussion using the Large Language Models (LLMs) and structured databases. This research is of great importance to our Gadget Recommendation System project as it gives a roadmap to incorporate LLAMA-3 with a MySQL database. With the integration of LLMs to perform detailed computing together with quick database search by our system, our system can meet demands and requests for complicated queries, accurate gadget information search, and customized recommendations for users, thereby enhancing the satisfaction rate of the users utilizing the system and the efficiency of the system. The article's findings in query synthesis and execution are directly relevant to how we translate user inputs that are in natural language into the required database queries for proper gadget recommendations.

3. Improving RAG Quality for Large Language Models with Topic-Enhanced Reranking [3]: This paper describes a conception of how recommendation systems that use artificial intelligence and natural language processing are used to provide users with personalized suggestions for an improved experience. The paper deals with many models of recommendation, especially hybrid approaches that bring structured data with semantic analyses to deal with complex queries better than other recommendation measures normally do. This applies to our Gadget Recommendation System, in which we wish to use LLAMA-3 along with structured data from MySQL to give personal gadget recommendations. This hybrid approach should pave the ground for a more intuitive contextual recommendation system along the lines of our project's goals of increasing user satisfaction and support in decision-making.

4. Large Language Models: A Survey [4]: The article discusses the use of Large language models (LLMs) in natural language understanding and how such skills are incorporated into recommendation systems for contextual accuracy and relevance. Also, it discusses how LLMs treat complicated user queries and use extensive training data to produce contextually relevant responses, which help engage the user with more accurate and relevant recommendations. Combining structured data with LLM-based semantic reasoning, as we did with LLama 3 and MySQL DB, indeed takes the ballpoints for efficient information processing, crafting out accurate SQL queries, and personalized gadget recommendations from much-varied user queries.

5. Generating Personalized Recommendations via Large Language Models (LLMs) [5]: This article outlines that LLMs can complement the recommendation procedures by increasing their accuracy and relevance to a user's preferences. Strengthened by the results of the analysis of natural language, the article explains how LLMs provide a deeper focus on user preferences than traditional recommendation systems and, therefore, how systems can generate more varied and accurate recommendations. This article directly aligns with our Gadget Recommendation System since it demonstrates the integration of LLAMA-3 with structuring data to polish the query answers and give particular gadget recommendations. By doing so, our project will enhance user satisfaction and interaction by getting informed gadget recommendations that are closer to the user's needs and preferences.

6. Retrieval-Augmented Generation for AI-Generated Content: A Survey [6]: AI-generated content is a field that has changed with the use of other models such as Retrieval Augmented Generation (RAG). Several state-of-the-art generator models that are non-adversarial and based on the transformer architecture, show quite decent performance in terms of generating natural and relevant context text. Current writings stress one major advantage of RAG, they show that it is better than merely generative models in the task that may involve some domain or dynamic knowledge. Including retrieval, procedures makes these models context-sensitive and flexible, valuable when used in question-answering systems, summarization, and advanced dialogical interfaces. This means that retrieval latency, integration complexity, and the quality of the retrieved data will be the key areas of consideration for future research as RAG seeks to refine the effectiveness and efficiency of its information-retrieval processes.

7. The article "Retrieval-Augmented Generation for Natural Language Processing: A Survey" [7] techniques combining retrieval methods and generative language models in general. RAG has an information retrieval component that fetches relevant information; the language model uses this information to generate responses that are constrained in a context around what the user has asked. This takes care of the shortcomings of the LLM by responding on the basis of such factual retrievable knowledge, which aids the improvement of precision as well as coherence for LLM. The insights gained from this article helped our project, while we use Table-Augmented Generation (TAG) for querying into the database, RAG principles further solidify the rationale behind integrating retrieval approaches to allow LLaMA to provide accurate gadget information. Our system, by applying RAG's structured doctrine, will be able to get more accurate answers by retrieving with respect to specific gadget detail that fits into user queries for further relevance and satisfaction of the recommendations.

8. TechCritique: Intelligent Gadget Review and Recommen-

dation System [8]: The article tells about a gadget review platform designed to fully realize the user-centered experience by encompassing several modular options, including a comparison tool, price tracker, chatbot, recommendations engine, and real-time news updates. Thus, the multi-functional system can address constant challenges in the gadget market while personally recommending a wide-range, credible, up-to-date insight for helping such tech enthusiasts make informed decisions. This article relates to our Gadget Recommendation System in that both systems aim to improve the user experience through personalized gadget recommendations. The modularity discussed with reference to TechCritique lays an interesting context in which data-optimized user experience dovetails with our design principles involving real-time user input and structured gadget data to deliver accurate and personalized gadget recommendations.

9. Llm-rec: Personalized recommendation via prompting large language models [9]: The article deals with advances in text-based recommendation systems, highlighting some of the challenges facing alignment with user preferences caused by incomplete descriptions of items favoring accuracy. The recent approaches have employed knowledge bases and pre-trained language models for contextual enrichment, large language models (LLMs) have come under recognition for their potential to fill these gaps because of their wide world knowledge and reasoning capabilities. recommendation-driven, and engagement-guided-and their combinations. This new system seeks to increase performance in the recommendation process with enriched input text and performance either on par with or superior to other complicated content-based models proving their usability for personalized recommendation with little supervision by experts or domain knowledge.

10. Llama 3.1: An In-Depth Analysis of the Next Generation Large Language Model [10]: In this article, Llama 3.1, an AI foundation model, has been introduced and explained in terms of its advanced capabilities. Llama 3.1 retains the flagship model with 405 billion parameters, besides 8B and 70B versions designed for a range of applications including but not limited to multilingual translation, coding, reasoning, and tool use. It comes with a context length of 128K tokens, enabling the processing of long input sequences. This article is very helpful to our Gadget Recommendation System Project since it illustrates Llama 3.1's strong reasoning and fine tool integration abilities, both very critical to interpreting user queries and producing adequately accurate recommendations that observe the "world of context". The great multilingual and reasoning functions of the model assist in raising the overall level of performance in dealing with diverse and complex queries posed by the user in providing accurate customized recommendations about gadgets.

III. Objectives of the Study

- To develop a recommendation system using LLaMA-3 8 Billion to natural language processing to effectively interpret user queries, generating contextually accurate and relevant gadget suggestions based on user requirements.
- To integrate LLaMA-3 8B with Table-Augmented Generation to give structured MySQL database access and enhance the system's ability to handle unstructured user queries while effectively reflecting on gadget recommendations.
- To achieve an initial 40-60% accuracy in generating gadget recommendations. The increased number of attempts in processing user queries leads to noticeable improvements in the system's accuracy reflecting its potential to iteratively refine responses and advance extremely precise and user-relevant recommendations.

IV. Data Collection

For the Gadget Recommendation System, two kinds of datasets are used, mobile devices and laptops. These datasets were sourced from reliable online platforms and well-cleaned to ensure data quality and relevance to this project's goals. These datasets provide a detailed view of mobile and laptop product specifications, which supports us in making recommendations for Gadgets.

Mobiles Dataset:
The mobile dataset was scraped from the GSM Arena website. Hence generated data was cleaned and filled missing values using average values for numeric fields and relevant information for categorical fields. This dataset has three categories of data, which includes mobiles, Tablets and smartwatches. Additionally, a few columns were added or removed to align with this project requirements and the specifications. This dataset includes a wide range of attributes, capturing detailed specifications for each mobile device. Device attributes include:

1) **Mobile ID**: Unique identifier and Primary Key for each mobile device.
2) **Model Name**: The name of the mobile model.
3) **Brand**: The brand of the mobile device.
4) **Category**: Device type (Mobile, Tab, Smartwatch).
5) **Network Support**: Indicates support for 2G, 3G, 4G, and 5G.
6) **Announcement and Release Details**: Information on the announcement and release dates, as well as the device's status.
7) **Dimensions and Weight**: Physical dimensions (in mm and inches) and weight.
8) **Display Specifications**: This includes display type, size (in inches and cm), screen-to-body ratio, resolution (in pixels, ratio, and PPI), display protection, and other display features.
9) **OS and Hardware**: Operating system, chipset, CPU, GPU, memory details, RAM, external memory, and memory slot are covered.
10) **Camera Details**: Specifications of the main and front cameras, including resolution, features, and video capabilities.
11) **Connectivity**: WiFi support, Bluetooth, GPS, NFC, radio, USB, and sensor details.

12) **Battery and Charging**: Battery type and charging specifications.
13) **Other Features**: Color options, model variations, SAR rating, AnTuTu scores, benchmark results, popularity, rating, and price in USD.
14) **AI Features**: Special AI-driven functionalities.

Laptop Dataset:
The laptop dataset was sourced from Kaggle and then cleaned it to meet the research needs. Missing values were filled with relevant data, and prices, originally in Euros, were converted to USD. Columns were streamlined to meet the research needs.

1) **laptop ID:** Unique identifier for each laptop.
2) **Brand:** Laptop brand.
3) **Name and Model:** The name and model of the laptop.
4) **Screen Size:** Screen size specification.
5) **Operating System (OS):** The operating system installed on the laptop.
6) **Price:** Price in USD.
7) **Processor:** Details on processor name, brand, and speed (GHz).
8) **RAM Details:** Information on RAM size, type, and expandability.
9) **Display Type:** Type and quality of the display.
10) **Graphics:** GPU specifications and brand.
11) **Storage:** SSD and HDD storage capacity.
12) **Power and Battery Life:** Adapter details and battery life.

By structuring and managing these datasets effectively, it is ensured that the data is both comprehensive and relevant for analysis. The clean and clear data hence generated will lay a strong foundation for generating efficient results for this research.

A. Exploratory Data Analysis(EDA)

EDA plays an important role in understanding the structure and trends within the dataset. The datasets used in the gadget recommendation system consist of two main collections of device data, mobiles and laptops. The mobile dataset consists of 559 rows and 65 columns, where each row represents a unique mobile device. Each column gives certain features about the brand, model, screen size, processor, RAM, battery, and price. The large number of columns ensures that sufficient information is available for effective recommendations. The laptop dataset contains 3,976 Rows and 22 Columns; each row represents one particular laptop. It includes key specifications like the brand, operating system, processor, RAM, storage, display type, and price. Below are the price statistics for the data used in the gadget recommendations system. The main observations from this statistics table are prices ranging from $95.88 to $6046.68, showing a wide variety of laptops from budget to premium categories. The mean price is $869.19, with a standard deviation of $626.49, indicating variability in pricing.

First, we preprocessed the data by filling in missing values and erasing duplicates to ensure consistency in our data. Missing values were dealt with either by filling them with appropriate measures, such as the mean or median, or by removing rows or columns containing missing data. Data types were also checked and corrected where necessary to meet the requirements of the dataset. The key insights from the visualizations were patterns related to the distribution of key specifications, such as RAM, processor types, and price ranges. Other interesting visualizations included brand-wise product counts and feature comparisons that shed light on market trends. For instance, fig 1 shows the distribution count for the different devices(mobiles, tablets, and smartwatches) provided by various brands. Motorola and Samsung have the highest number of devices with the contribution of each at approximately 140 units, representing how strong they are in the market.

TABLE I
DESCRIPTIVE STATISTICS FOR PRICE

Statistic	Price
Count	3976
Mean	869.19
Std Dev	626.49
Min	95.88
25th Percentile	478.48
Median (50%)	707.88
75th Percentile	1019.88
Max	6046.68

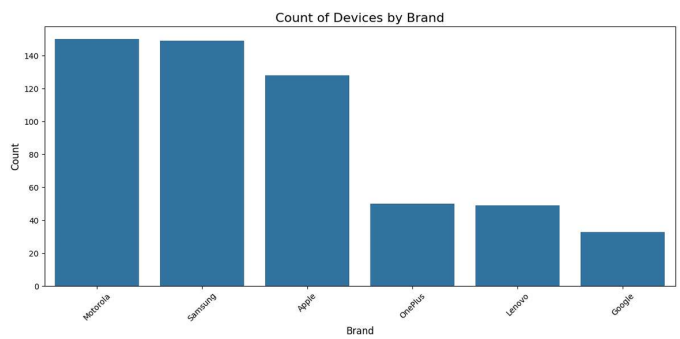

Fig. 1. Count of Mobiles by Brand

Fig 2 shows the distribution of laptops across brands, highlighting which brands dominate the market in terms of the number of available models. The most represented brand has a significantly higher number of laptops compared to other brands, indicating its popularity and market reach.

B. Data Visualization

The following visualizations show the distribution of different parameters in the dataset used for the project. Fig.3, shows the price distribution of devices(mobiles, tablets, and smartwatches) across various brands in USD. These visualizations will help in analyzing pricing trends and variations among brands to draw inferences on their market segmentation and target audiences.

Fig.4 shows the distribution of laptops using processors from three major brands: AMD, Intel, and MediaTek. Each section of the pie chart represents the proportion of laptops

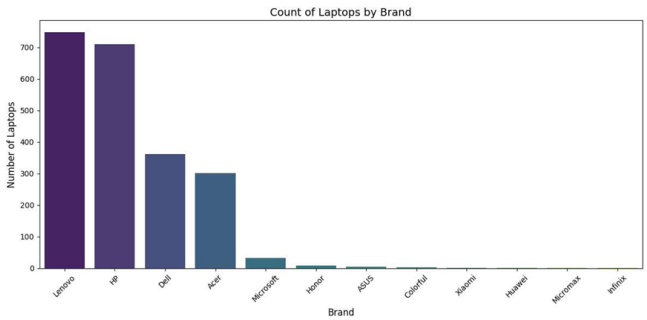

Fig. 2. Count of Laptops by Brand

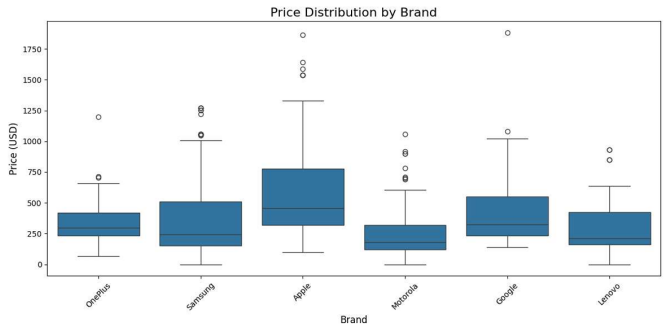

Fig. 3. Price Distribution by Brand

featuring processors from a particular brand. Intel leads the dataset in processor contribution, followed by AMD and then MediaTek.

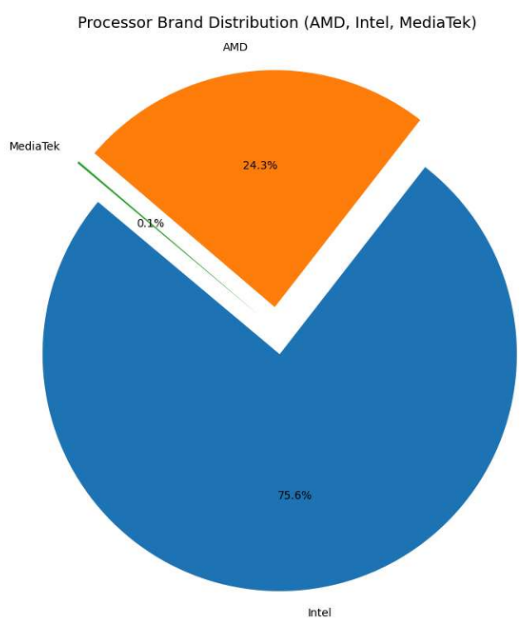

Fig. 4. Processor brand distribution

C. Data Analytics

For the analysis techniques in our project, we used Python for the analysis of data. A correlation matrix was utilized to show the relationship between key features price, RAM, battery life, SSD, and adapter power-finding insights into feature inter-dependencies that influence gadget recommendations. This correlation matrix visualizes the relationship between the key variables of the dataset: Price, RAM, Battery Life, SSD, and Adapter. Each cell shows the correlation coefficient between two variables, moving from -1 (perfect negative correlation) to 1 (perfect positive correlation). The

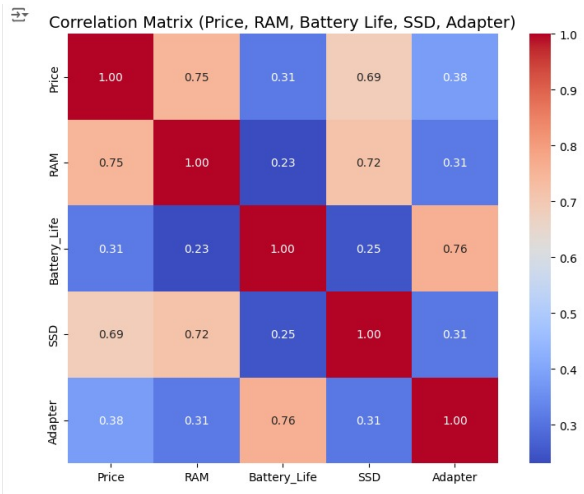

Fig. 5. Correlation Matrix

key insights from the figure are:

- Price and RAM are strongly positively correlated at 0.75, indicating that devices with higher RAM are usually more expensive.
- Price and SSD also have a high positive correlation of 0.69, which means that the device price increases with increased SSD capacity.
- Battery Life and Adapter have a moderate positive correlation of 0.76, which may indicate that better battery life could mean higher powerful adapters.
- RAM and SSD are highly correlated at 0.72, which means that with increased RAM, the SSD storage is also increased in most devices.
- Battery Life has a weaker correlation with other variables except for the adapter, reflecting its less direct influence on price or memory-related components.

V. System Architecture/Modeling approaches

Our project utilizes structured information and Large Language Models(LLMs) to give the output to the user and provide a wide range of gadget recommendations. This system architecture applies methodologies for query synthesis, structural data manipulation, and natural language generation for the recommendation mechanism by multiple users' requirements.

User Query Interface: The user has to enter their gadget-related questions via a Django-based interface. The natural language query captures things like the preference for gadgets, specifications, budget specific, etc to the user. The user request (R) is reset with an initial value, and an attempt count has been

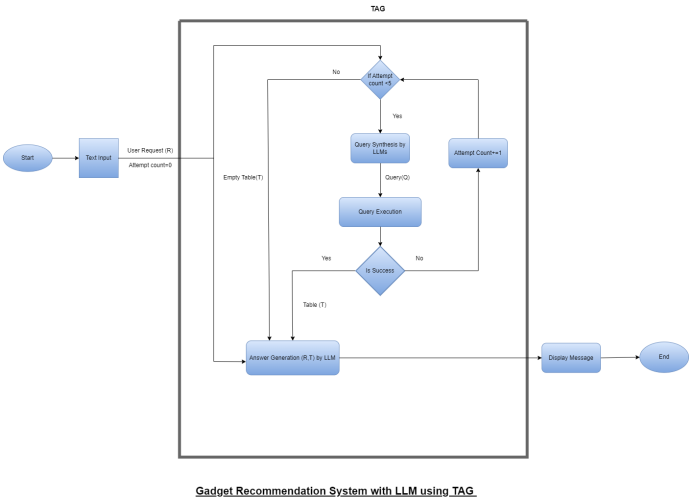

Fig. 6. Flowchart

set to 0. This helps track the number of times the system tries to process the query.

Query Synthesis: Using the TAG method, the user's natural language query R is transformed into a structured SQL query Q . The synthesis process translates text input of almost any complexity level and lack of structure into a more or less structured SQL query to be executed against the back-end database. Thus, this step employs the LLM to create an SQL query so that it aligns with the functional requirements expected by the user. After each attempt, the Attempt count is incremented by 1 to keep track of retries in failed attempts. Each time the query execution fails, LLM regenerates the query. This recursion halts after 5 attempts and sends an empty data frame for answer generation.

Query Execution: In this stage, the synthesized SQL query (Q) is executed on a MySQL database containing structured gadget data like gadget specifications, prices, and feature information. This process extracts some relevant data, which is then organized into a structured table (T) for further processing.

Response Generation with LLaMA-3 8B: In the answer generation stage, LLaMA-3 8 Billion processes the user's original query (R) and the obtained table (T) and generates an answer (A) that is personalized. In this way, LLaMA-3 8Billion proposes a better recommendation with a clear understanding. The results table stores every user question, Generated SQL query, query execution response, LLM response, and attempt count for accuracy measurement.

Display Message: In this stage, the system shows the generated answers to the user, thus completing the recommendation process.

Model Evaluation:-Exact Match Method

For determining the performance accuracy of our Gadget Recommendation System, we used the Exact-Match Method a technique for obtaining the correctness of the system's responses. We manually created a set of 53 questions that cover a wide range of gadget-related queries based on the data in our database. Each question was assigned correct answers based on our data as references for evaluating the system's output. We then asked the system the same questions. The accuracy is calculated according to the suggestions provided by the questions passed to the model. For each response generated by the system, we checked whether it matched the manually prepared correct answer or not. A response was considered "correct" if it matched the expectation. Otherwise, the answer is considered as "incorrect." The accuracy of the model was calculated using the following formula:

$$\text{Accuracy}(\%) = \left(\frac{\text{Number of Correct Answers}}{\text{Total Number of Questions}}\right) \times 100$$

Using this approach, we can evaluate how often the system produces the exact expected answer. By manually comparing the expected and actual answers, we can identify specific areas where the system needs to improve query synthesis or response generation. The system achieved an 80% Exact Match Accuracy. This means that 80% of the questioned user queries converted into SQL matched exactly with the expected output.

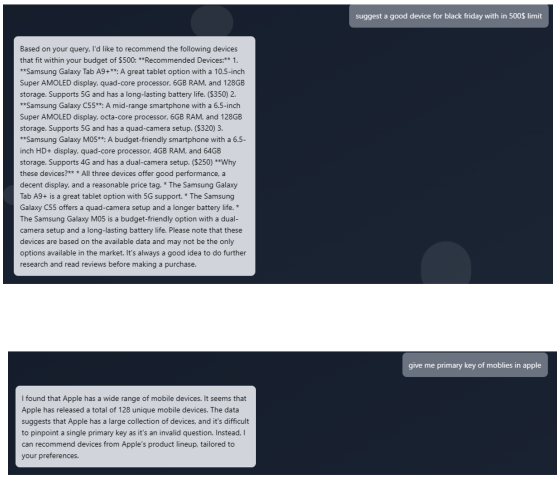

Fig. 7. Sample Results

VI. Hypothesis

The hypothesis is whether integrating large language models (LLMs) such as Llama-3 8 Billion with structured data may improve the operation of gadget recommendation systems over rule-based or filtering techniques. Current text-to-SQL methods achieve only around 20% accuracy, in our project using the TAG method, we predict noticeable improvements and reach about 40% accuracy. This hypothesis suggests that, due to their efficiency in natural language processing, LLMs will parse users' questions with more contextual meaning and ultimately yield customized recommendations. The ability of the LLM to parse complex and unstructured queries combined with structured data retrieval from databases will provide a gadget recommendation that is more accurate and relevant to the user. This model better translates different user inputs into

SQL query meanings while presenting personalized recommendations to the specific user. This work assists in evaluating how an LLM model compares with the conventional design to identify how it improves the accuracy of recommendations and user satisfaction across different categories of electronic devices.

VII. Results

The exact match method is most commonly used in recommendation systems to measure accuracy. It checks how frequently the output of a model matches the expected answer or result precisely, without any approximation or partial correctness allowed. For evaluation, a Results table has been created in the database, which includes fields like user_question, LLM_SQL, LLM_response, and execute_mysql_query_error. A set of 53 questions has been generated, manually, to test the TAG methodology. These questions are generated completely based on the dataset used. Each query is assessed by manually verifying the generated SQL output and LLM response.

The primary goal of this table is to capture:

- The initial question from the user
- The SQL query generated by the system
- The response provided based on the generated query
- Any errors encountered during query execution

Row Labels	Attempt count=1		Attempt count=3		Total No. of manually verifed Queries	Total No. of Execution errors
	No. of manually verifed Queries	No. of Execution errors	No. of manually verifed Queries	No. of Execution errors		
Wrong Answers	16	11	7	4	23	15
Exact Match	37	0	46	0	83	0
Grand Total	**53**	**11**	**53**	**4**	**106**	**15**
Percentage of Accuracy	**69.81**		**86.79**			

Fig. 8. Performance results in terms of accuracy at various attempts

In our Gadget Recommendation System, the exact match method evaluates how accurately the system responds to user queries. For example, when a user asks for a specific gadget feature or specification, the system's output would be considered right only if it agrees word for word with the predefined expected answer. We manually generated around 53 questions based on the data in our database. We passed these questions to our model and obtained results. To observe the accuracy and improvement over different iterations, we allowed 5 attempts for each question, with all the responses being logged in the database for analysis. For the 1st attempt, 42 out of 53 were correct, resulting in around 69% accuracy. For 3rd attempt, the accuracy increased to 80% with the number of incorrect answers considerably reduced. This indicates the system's ability to improve the quality of answers over multiple attempts.

Although multiple attempts increase processing time, this approach is useful when accuracy is prioritized. This method of iterative evaluation provides a validation mechanism to show improvement in performance due to repeated attempts.

VIII. Future Research

There is an enormous scope for further improvements in our project, either in the functionality or usability of the system. Increasing the number of regeneration attempts for a given SQL query will boost the quality of recommendations when the user queries are complex or vague. On the other hand, the market for every kind of gadget must provide live and vast data that is to be incorporated to help the system stay relevant and thus make it highly applicable in the real world. Implementing user sessions can help personalize users' experiences by storing and using their previous queries. This feature can alert when their gadget becomes available on the market, along with recommendations specifically for these users, increasing their satisfaction and active engagement.

IX. Conclusion

By the end of the project, our Gadget Recommendation System was 80% accurate in generating exact SQL matches, showing strong performance for most of the user questions. This precision shows the effectiveness of combining LLaMA-3 8 billion with structured data to provide precise recommendations. We have also observed that with increasing attempts, the accuracy steadily improved, allowing the system to refine its query synthesis in every iteration and give much better recommendations on subsequent tries thus increasing the reliability of responses. This strategy turned out to be beneficial in solving complex queries and hence obtaining high-accuracy results.

References

[1] A. Biswal, L. Patel, S. Jha, A. Kamsetty, S. Liu, J. E. Gonzalez, C. Guestrin, and M. Zaharia, "Text2sql is not enough: Unifying AI and databases with tag," *arXiv preprint arXiv:2408.14717*, Aug. 27, 2024.

[2] A. Purwar, "Evaluating the efficacy of open-source LLMs in enterprise-specific RAG systems: A comparative study of performance and scalability," *arXiv preprint arXiv:2406.11424*, Jun. 17, 2024.

[3] N. Ampazis, "Improving RAG Quality for Large Language Models with Topic-Enhanced Reranking," in *IFIP International Conference on Artificial Intelligence Applications and Innovations*, Springer, 2024, pp. 74–87.

[4] S. Minaee, T. Mikolov, N. Nikzad, M. Chenaghlu, R. Socher, X. Amatriain, and J. Gao, "Large language models: A survey," *arXiv preprint arXiv:2402.06196*, Feb. 9, 2024.

[5] H. Sidahmed, S. Ahuja, and M. K. Tiwari, "Generating personalized recommendations via large language models (LLMs)," 2022.

[6] P. Zhao, H. Zhang, Q. Yu, Z. Wang, Y. Geng, F. Fu, L. Yang, W. Zhang, and B. Cui, "Retrieval-augmented generation for AI-generated content: A survey," *arXiv preprint arXiv:2402.19473*, Feb. 29, 2024.

[7] S. Wu, Y. Xiong, Y. Cui, H. Wu, C. Chen, Y. Yuan, L. Huang, X. Liu, T. W. Kuo, N. Guan, and C. J. Xue, "Retrieval-augmented generation for natural language processing: A survey," *arXiv preprint arXiv:2407.13193*, Jul. 18, 2024.

[8] K. Y. Shinde, R. Y. Boraste, A. N. Shaikh, and P. B. Khairnar, "Tech-Critique: Intelligent Gadget Review and Recommendation System," May 2024.

[9] H. Lyu, S. Jiang, H. Zeng, Y. Xia, Q. Wang, S. Zhang, R. Chen, C. Leung, J. Tang, and J. Luo, "LLM-Rec: Personalized recommendation via prompting large language models," *arXiv preprint arXiv:2307.15780*, Jul. 24, 2023.

[10] R. Vavekanand and K. Sam, "Llama 3.1: An in-depth analysis of the next-generation large language model," 2024.